TSM 初学者教学版 (中文) v20261015-0757

① 一句话

Taiwan Semiconductor Manufacturing Company Limited 是一家 Foundry / Advanced packaging 晶圆代工/先进封装 公司。现在股价 $430.00。我们的 Base 模型认为每股价值约 $350（-18.6%）。

② 发生了什么

最近一季财报 EPS 实际 vs 预期: [‘2026-06-30’, ‘4.31’, ‘3.88658’, ‘0.1089’]（实际、预期、超预期%）。下次财报约 2026-10-15。

③ 术语 Glossary

- DCF (Discounted Cash Flow｜现金流折现): 把未来每年能拿到的现金，按”时间成本”折回今天加总。
- FCF (Free Cash Flow｜自由现金流) = 经营现金流 − 资本开支。
- WACC (Weighted Average Cost of Capital｜加权平均资本成本): 投资人要求的年回报率，用作折现率。
- Reverse DCF (反向 DCF): 反过来问”现价要求公司增长多快”。
- Consensus (一致预期): 分析师预测的平均值。
- Falsifier (证伪指标): 一旦出现就说明论点错了的信号。

④ 生活比喻

买这家公司股票≈买一棵果树。DCF 就是估算果树未来每年结多少果(FCF)，再考虑”明年的一斤果不如今天的一斤果值钱”(WACC 折现)。

⑤ 因果箭头

Foundry / Advanced packaging 晶圆代工/先进封装 需求 → 收入增长 (g) → 自由现金流 (FCF = 收入 × FCF 率) → 折现 (÷(1+WACC)^t) → 每股价值 → 与股价比较 → 预期差

⑥ 数字例题（符号定义）

设 R_0 = 过去 12 个月收入 = 4,440.5 bn TWD；g_1 = 第一年增速 = 35.9%；m = 长期 FCF 率 = 30.0%；W = WACC = 11.4%。 第一年收入 $R_1 = R_0 \times (1+g_1)$ = 6,034.6 bn TWD。 若第 5 年 FCF 率达到 m，则第 1 年 FCF ≈ $R_1 \times m_1$（m_1 由当前 25.3% 向 30.0%

线性过渡）= 1,584.8 bn TWD； 折现到今天: $FCF_1 \div (1+W)^1$ = 1,422.6 bn TWD。十年加总 + 终值 = EV 55,999.5 bn TWD。

⑦ 事实 / 预测 / 市场预期

- 事实: 过去财务数据(第 4 节)、现价。
- 预测(我们的假设): 长期 FCF 率 30.0%、三情景。
- 市场预期: 一致预期增速 35.9%；价格隐含增速 41.9%。

⑧ 三情景

Bear $134 / Base $350 / Bull $655；概率加权 $372。

⑨ 反方观点与失效条件

反方: Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额 失效条件: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修

自我检验题

1. 如果 WACC 上升 1 个百分点，价值会升还是降？为什么？（提示：分母变大）
2. 反向 DCF 隐含增速高于一致预期意味着什么？

TSM Beginner Teaching Edition (English) v20261015-0757

① In one line

Taiwan Semiconductor Manufacturing Company Limited is a Foundry / Advanced packaging 晶圆代工/先进封装 company. Price $430.00; our Base model value ≈ $350 (-18.6%). ## ② What happened Last quarter EPS actual vs consensus: [‘2026-06-30’, ‘4.31’, ‘3.88658’, ‘0.1089’]. Next earnings ≈ 2026-10-15 (TENTATIVE). ## ③ Glossary DCF (Discounted Cash Flow｜现金流折现); FCF (Free Cash Flow｜自由现金流) = operating cash flow − capex; WACC (Weighted Average Cost of Capital｜加权平均资本成本) = the yearly return investors demand; Reverse DCF (反向 DCF) = what growth the price requires; Consensus (一致预期); Falsifier (证伪指标). ## ④ Analogy Owning the stock ≈ owning a fruit tree: DCF

estimates future fruit (FCF) and discounts it because fruit next year is worth less than fruit today. ## ⑤ Causal arrows Demand → revenue growth (g) → FCF (= revenue × FCF margin) → discounting ÷(1+WACC)^t → value/share → compare with price → expectation gap ## ⑥ Worked example (symbols) R_0 = TTM revenue = 4,440.5 bn TWD; g_1 = 35.9%; m = long-run FCF margin = 30.0%; W = 11.4%. $R_1 = R_0 \times (1+g_1)$ = 6,034.6 bn TWD; FCF_1 = 1,584.8 bn TWD; PV = $FCF_1/(1+W)$ = 1,422.6 bn TWD; 10 years + terminal → EV 55,999.5 bn TWD. ## ⑦ Fact vs forecast vs market expectation Facts: reported financials, price. Forecast: our margin/scenario assumptions. Market: consensus growth 35.9%; price-implied growth 41.9%. ## ⑧ Three scenarios Bear $134 / Base $350 / Bull $655; weighted $372. ## ⑨ Counterview & failure conditions Geopolitical shock, US tariffs on chips, Intel/Samsung regain share｜地缘冲击、美国芯片关税、英特尔/三星夺回份额 — Falsifiers: GM falls below long-term target band; monthly revenue y/y decelerates sharply; capex cut｜毛利率跌破长期目标区间；月营收同比骤降；Capex 下修 ## Self-check 1. WACC +1pp → value up or down? 2. What does price-implied growth > consensus mean?

TSM — Research logic map｜研究逻辑图



Reverse DCF 反向DCF: price implies Y1 growth ≈ 42%

Consensus FY+1 growth 一致预期: 36% → gap = expectation the price embeds 价格隐含预期差

Claim class 声明类别: INFERENCE / HYPOTHESIS | Source 来源: Profile (analyst framework) + model
Thinking 思考: Read left→right: each box must have evidence; the falsifier box tells you what would prove the thesis wrong. | Not investment advice 非投资建议

logic